

PEAK ACHIEVERS ONLINE ACADEMY

GRADE 7 INTEGRATED SCIENCE

REVISION BOOK - BOOK 1

Learner-Friendly CBE/CBC Edition • Practical Science • Application Quizzes • Mixed Concepts • CBE Assessment Test Models

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LEARN • APPLY • THINK BIG • ACHIEVE

FREE SAMPLE
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This is an independent educational revision resource. Its assessment models are original Peak Achievers materials and are not official KNEC/KICD examination papers.

Dedication

In loving memory of my late father, Simeon Nyatuka, whose guidance, sacrifice and inspiration remain part of my educational journey.

With heartfelt appreciation to **Hyrine Kerubo, Peter Munde Mong'are, Teresa Kwamboka Sawe, Mnaree, Richie, Duke, Caleb, Cuba, Philo, Rob, Tim, Mr Peter Kimani**, and all those whose support and encouragement have strengthened this journey.

To Her, my special one, for being a special part of my journey and for the place you hold in my heart.

To my fellow teachers, guardians, family and every learner who continues to inspire the work behind Peak Achievers.

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How to Use This Book

Start with the simple explanation. Study the illustration. Carry out the practical activity with teacher guidance. Read the Peak Achievers boxes. Attempt the Application Quiz before checking its answer. Finish with the Revision Exercise. The final section contains mixed-concept tests and CBE Assessment Test Models.

LEARNING PATH

UNDERSTAND → OBSERVE → PRACTISE → INVESTIGATE → APPLY → INTERPRET → SOLVE

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1. Define Integrated Science.
2. Name the three broad branches.
3. Give four everyday uses of science.
4. Write the investigation cycle.
5. Explain why evidence matters.

1.2 Laboratory Safety

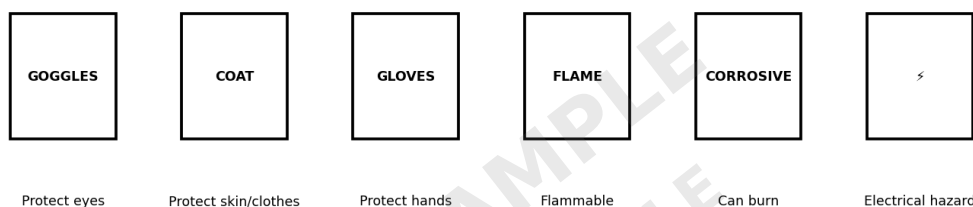
A laboratory is a place for practical learning. Safety rules help us investigate without avoidable harm.

Hazard and risk

A hazard can cause harm. A risk is the chance that the harm may happen. Broken glass is a hazard; being cut is a risk.

Safe practice

Follow instructions, wear suitable PPE, keep the area tidy, do not run/eat/taste substances, report spills and breakages, and wash hands after practical work.



Common protective equipment and hazard reminders.

Practical Activity

Aim: to identify hazards and safe actions

Materials: teacher-prepared pictures or written laboratory situations

Procedure: For each situation identify the hazard, risk and one safe action.

Observe/Record: Make a Hazard | Risk | Safe action table.

Conclusion: Recognising hazards helps prevent accidents.

CAUTION

Do not create hazards for the activity.

REMEMBER

HAZARD → RISK → PRECAUTION → SAFE ACTION

CAUTION

Never taste an unknown substance or directly smell an unknown chemical.

NB

Hot glass may look like cold glass.

THINK BIG

Risk changes with how a substance is handled.

EXAM SMART

State the exact safe action, not only 'be careful'.

Application Quiz - Try Before Checking

1. What should happen after a beaker breaks?

Answer: Keep others away, inform the teacher and use the approved collection method; do not use bare hands.

2. Why should an unlabelled bottle not be used?

Procedure: Measure the same sample three times at eye level.

Observe/Record: Record each reading with units.

Conclusion: Careful repeated readings improve confidence.

CAUTION

Wipe spills promptly.

REMEMBER

NAME → FUNCTION → CORRECT USE → UNIT

HINT

EYE LEVEL → MENISCUS → SCALE → READING → UNIT

CAUTION

Wrong eye position can cause parallax error.

THINK BIG

Check technique before blaming apparatus.

EXAM SMART

Name the apparatus and explain why it suits the task.

Application Quiz - Try Before Checking

1. Best apparatus for exactly 30 mL?

Answer: A measuring cylinder.

2. What is missing from a reading written only as 35?

Answer: A unit, such as mL.

3. Why may two readings differ?

Answer: Eye position, meniscus reading, movement/spillage or scale-reading differences.

Revision Exercise

1. State functions of five apparatus.
2. Define parallax error.
3. State 1 mL in cm^3 .
4. Choose instruments for mass/time/temperature/length.
5. Why is a beaker poor for an exact volume?

Practical Activity

Aim: to use litmus safely

Materials: teacher-approved labelled solutions, red/blue litmus, droppers

Procedure: Test small labelled samples with separate litmus pieces.

Observe/Record: Record Sample | Red litmus | Blue litmus | Conclusion.

Conclusion: Conclusions come from colour changes.

CAUTION

Never taste unknown solutions.

REMEMBER

Acid: BLUE → RED. Base: RED → BLUE.

HINT

Observation first, conclusion second.

CAUTION

Never taste an unknown solution.

SPOTLIGHT

CLAIM + EVIDENCE + REASONING.

THINK BIG

Poor crop growth can have many causes.

EXAM SMART

INDICATOR → OBSERVATION → CONCLUSION

Application Quiz - Try Before Checking

1. P turns blue litmus red. Classify P.

Answer: Acidic.

2. Q turns red litmus blue. Classify Q.

Answer: Basic.

3. What should a farmer do before treating suspected acidic soil?

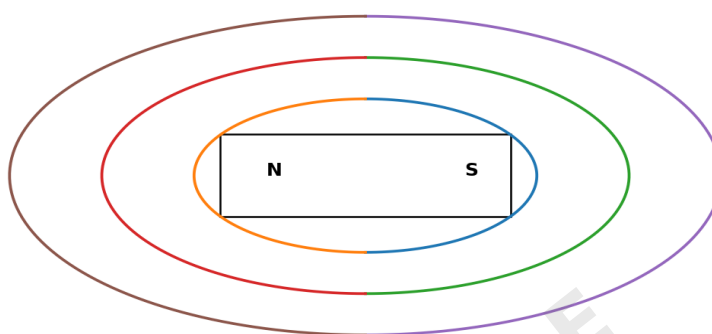
Answer: Test the soil and gather evidence.

Revision Exercise

1. Define indicator.
2. State litmus changes.
3. Give two familiar acidic substances.
4. State three safety rules.
5. Why record natural-indicator colours?
6. Define neutralisation simply.



Like poles repel; unlike poles attract.



Magnetic field around a bar magnet (schematic)

Simplified magnetic-field pattern.

Practical Activity

Aim: to identify magnetic materials

Materials: bar magnet and approved objects

Procedure: Bring the magnet close to each object and record attraction/no attraction.

Observe/Record: Make Object | Attracted? | Classification.

Conclusion: Not all metals are magnetic.

CAUTION

Keep magnets away from sensitive items as directed.

REMEMBER

LIKE REPEL; UNLIKE ATTRACT.

HINT

Repulsion is strong evidence when testing suspected magnets.

CAUTION

Do not write 'all metals are magnetic'.

NB

Nearby magnets can disturb a compass.

THINK BIG

Attraction can occur between a magnet and unmagnetised magnetic material.

EXAM SMART

OBSERVATION → PROPERTY → PRINCIPLE → CONCLUSION

Application Quiz - Try Before Checking

1. N near N?

Answer: They repel.

2. Does attraction of an iron nail prove the nail is a magnet?

Answer: No; iron can be attracted without being a permanent magnet.

3. How can magnets help recycling?

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